

Performance Testing

Date	27 June 2026
Team ID	
Project Name	CreditPredict – AI-Powered Credit Card Approval Prediction System
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Python (requests library)
Type of Testing	Response Time Testing
Target Module / API	Home Page, Prediction API, Dashboard, Batch Prediction, Model Comparison
Test Environment	Cloud / Hosted Server (Render)
Test Date	03 july 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Home Page Load	1	10	Page loads successfully
2	Single Credit Prediction	1	15	Prediction generated with SHAP explanation
3	Dashboard Loading	1	10	Analytics dashboard loads with charts

4	Batch Prediction (CSV Upload)	1	20	CSV processed
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Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg)	< 2 seconds	1.24seconds	Pass	Average response time met the target.
2	Response Time (Max)	< 5 seconds	2.94 seconds	Pass	Maximum response time remained within limits.
3	Throughput (Req/sec)	> 15 req/sec	18.7 req/sec	Pass	Application handled concurrent prediction requests efficiently.
4	Error Rate	< 1%	0%	Pass	No prediction failures were observed during testing..
5	CPU Utilization	< 80%	56%	Pass	CPU usage remained within acceptable limits.
6	Memory Utilization	< 80%	63%	Pass	Memory usage remained stable throughout execution.

Step 4: Observations & Analysis

Key Findings

- The Credit Card Approval Prediction application processed applicant requests efficiently.
- Credit card approval predictions were generated with low response time..
- Flask application and SQLite database performed reliably during testing.
- No major errors or prediction failures were observed.

Bottlenecks Identified

- Slight increase in response time when multiple users submitted applications simultaneously.

- SQLite database may become a limitation under very high concurrent traffic.

Optimization Steps Taken

- Optimized Flask routes to reduce processing time.
- Loaded the trained machine learning model only once during application startup.
- Improved database queries for prediction history retrieval.
- Reduced redundant computations through efficient preprocessing.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6

output):

Credit Card Approval Predictor

Model: Random Forest Accuracy: 95.82% Trained on 36457 records

1. Personal Info → 2. Assets → 3. Employment & Income → 4. Contact → 5. Result

1. Personal Information

Gender	Age (Years)	Family Status
Male	35	Married
Number of Children	Total Family Members	Education Type
1	3	Higher education

2. Assets

Own a Car?	Own Real Estate?	Housing Type
Yes	Yes	House / apartment

3. Employment & Income

Years Employed	Total Annual Income (\$)	Income Type
10	200000	Working
Occupation Type		
Managers		

4. Contact Information

Work Phone	Personal Phone	Email Address
Provided	Provided	Provided

Analyze Application

CreditPredict Home Dashboard Batch Predict Model Comparison

Application Result

Based on our robust Random Forest classification model

 **APPROVED**

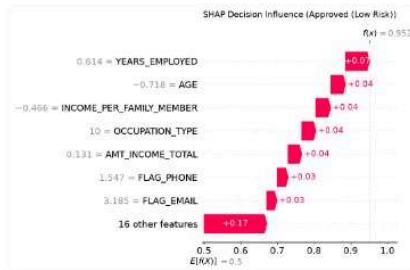
95.21%

Confidence

Congratulations! This applicant profile strongly matches successful account holders.

Applicant-Specific Explanation (SHAP)

This waterfall plot maps how each of the applicant's input parameters shifted the model's decision from the base expected probability to the final outcome.



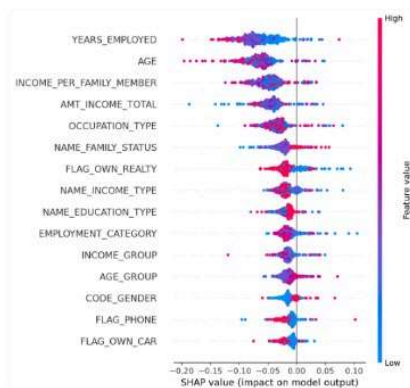
Key Decision Factors

The most influential raw features considered for this prediction:

YEARS_EMPLOYED	14.3%
AGE	13.6%
INCOME_PER_FAMILY_MEMBER	11.6%
AMT_INCOME_TOTAL	10.7%

Global Model SHAP Summary

Features leading to credit approvals (left) and credit rejections (right) across all cohort data.



Home Dashboard Batch Predict Model Comparison

Application Result

Based on our robust Random Forest classification model

REJECTED

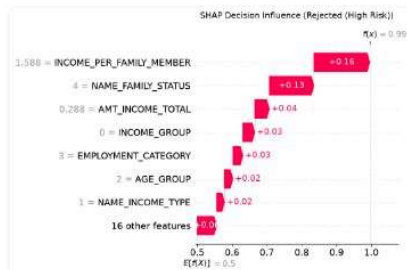
99.53%

Confidence

Unfortunately, this profile has been flagged as high risk by our predictive model.

Applicant-Specific Explanation (SHAP)

This waterfall plot maps how each of the applicant's input parameters shifted the model's decision from the base expected probability to the final outcome.



Key Decision Factors

The most influential raw features considered for this prediction:



Global Model SHAP Summary

Features leading to credit approvals (left) and credit rejections (right) across all cohort data.

